UNIVERSITY of HOUSTON | ECE

ECE 3337: Signals & Systems Analysis

Lecture 20: Fourier Analysis Techniques

Reference: Chapter 5
(Section 5.4 – 5.8)

Please turn off your gadgets now

How to Study for Final Exam

Date: May 11th 2 - 5pm, Room to be announced.

Plenty of old exams are posted on MS Teams for practice

- Practice for accuracy and speed
- Test yourself individually, rather than as a group
- Avoid a false sense of knowing a topic
  - This can easily happen in a group setting
- Suggested study strategy:
  - Do the practice tests individually
  - Identify the challenging problems, and ask your instructors for help
  - Discuss the challenging problems in your study group
  - Follow up with individual work to confirm that you understood the concepts correctly

Recap: Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)].$$

(sine/cosine representation) (5.27)

$$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t)\,dt$$

$$a_n = \frac{2}{T_0} \int_0^{T_0} x(t) \cos(n\omega_0 t)\,dt$$

$$b_n = \frac{2}{T_0} \int_0^{T_0} x(t) \sin(n\omega_0 t)\,dt$$

A BASIC "TOOL" FOR ENGINEERING

Adjustments to Fourier's Claim

A waveform $x(t)$ should satisfy some basic conditions for it to be expandable in terms of sinusoids

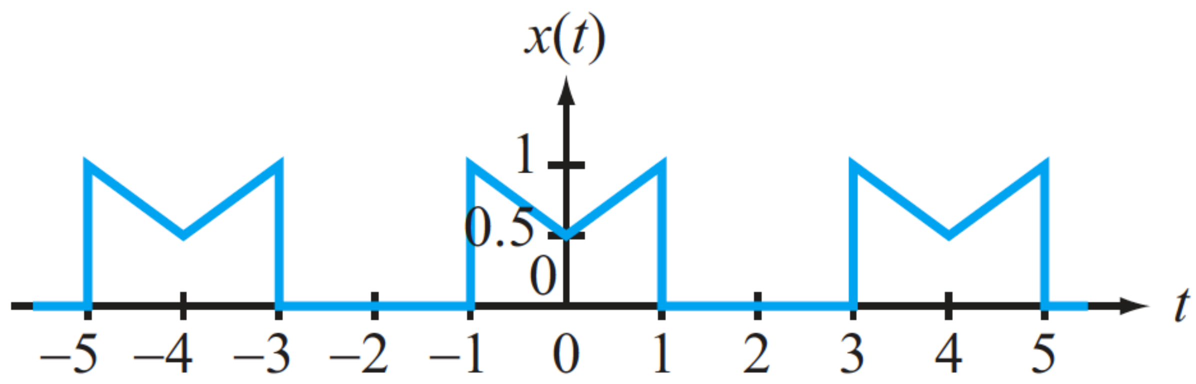
- It may only have a finite (not infinite!) number of discontinuities, maxima, and minima within one time period
- It must be "absolutely integrable":

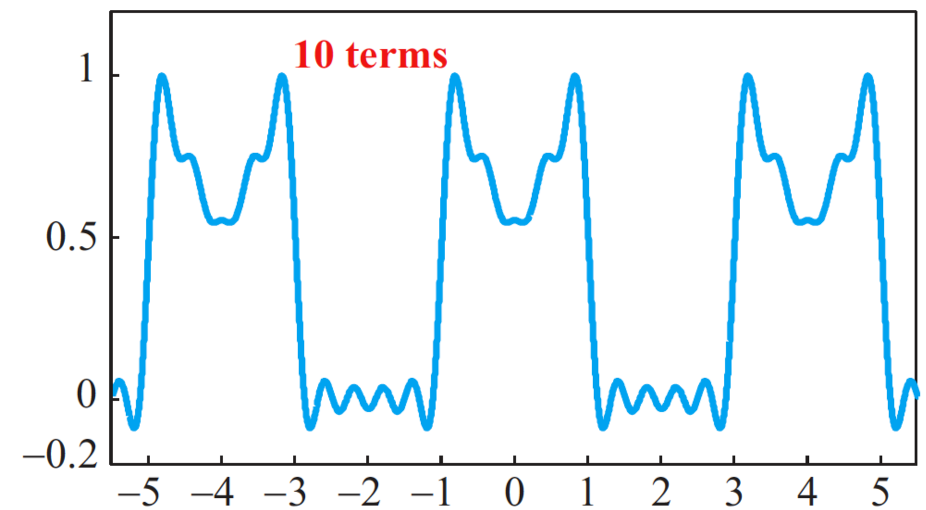
$$\int_0^{T_0} |x(t)|\, dt < \infty,$$

When these two conditions are met, the Fourier series converges to $x(t)$ in a *point-wise* manner *where there are no discontinuities*

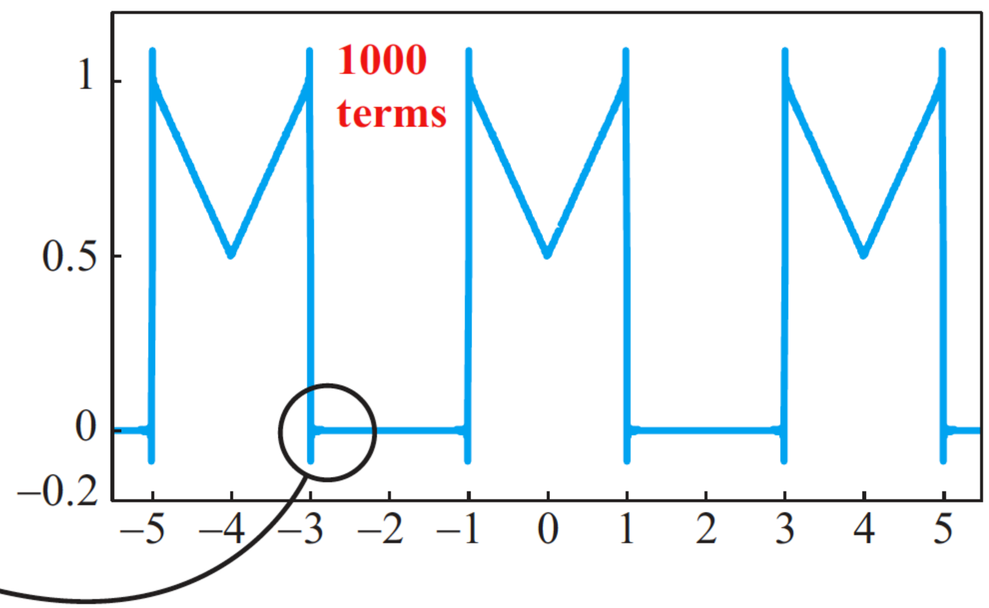
- We should still expect the Gibbs phenomenon at discontinuities

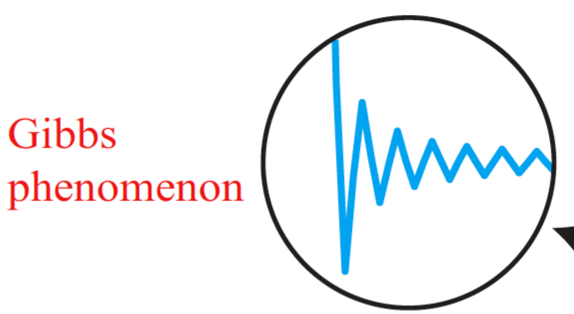
Discontinuities





ONLY GOES AWAY IN THE LIMIT $n \to \infty$





Today: Additional Forms of the Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]. \tag{5.27}$$

(sine/cosine representation)

✓

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n),$$

(amplitude/phase representation)

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t},$$

(exponential representation)

TODAY

#2 Amplitude & Phase Version

We need a concise version of the Fourier series for analyzing circuits using phasors

$$a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = c_n \cos(n\omega_0 t + \phi_n)$$

where:

$c_n =$ Amplitude of the n^{th} harmonic

$\phi_n =$ Phase of the n^{th} harmonic

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n),$$

(amplitude/phase representation)

Formula for c_n

$$a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t) = c_n \cos(n\omega_0 t + \phi_n)$$

Now,

$$\cos(x+y) = \cos x \cos y - \sin x \sin y$$

$$\Rightarrow c_n \cos(n\omega_0 t + \phi_n) = c_n \cos\phi_n \cos(n\omega_0 t) - c_n \sin\phi_n \sin(n\omega_0 t)$$

BY EQUATING THE CORRESPONDING TERMS, WE GET:

$$a_n = c_n \cos\phi_n$$

$$b_n = -c_n \sin\phi_n$$

$$\Rightarrow a_n - jb_n = c_n(\cos\phi_n + j\sin\phi_n) = c_n e^{j\phi_n}$$

THIS IS THE POLAR REPRESENTATION FOR c_n

Formula for c_n

$$a_n = c_n \cos\phi_n$$

$$b_n = -c_n \sin\phi_n$$

$$\Rightarrow a_n - jb_n = c_n(\cos\phi_n + j\sin\phi_n) = c_n e^{j\phi_n}$$

or

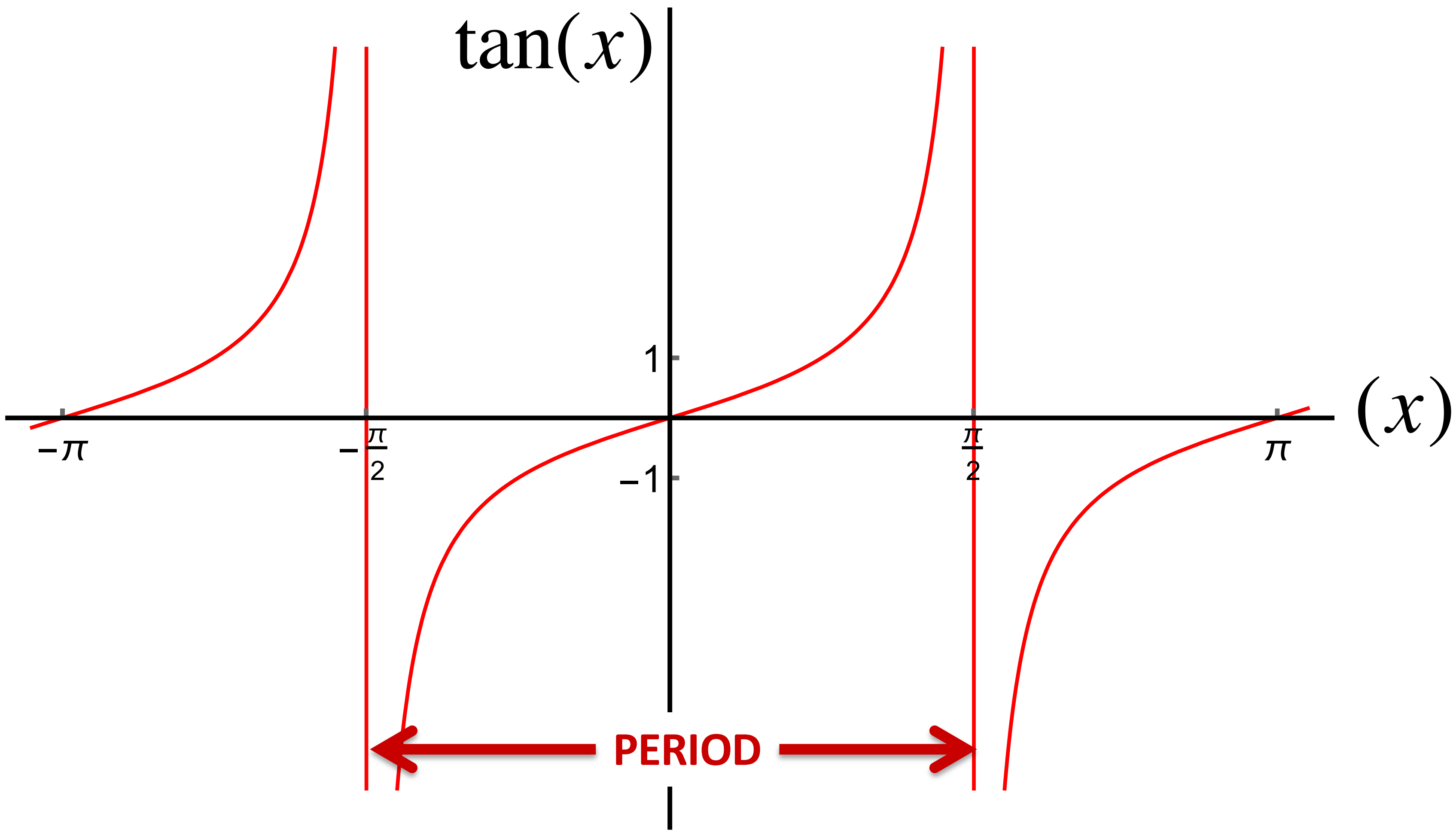
$$c_n = |a_n - jb_n| = \sqrt{a_n^2 + b_n^2}$$

$$\phi_n = \angle(a_n - jb_n)$$

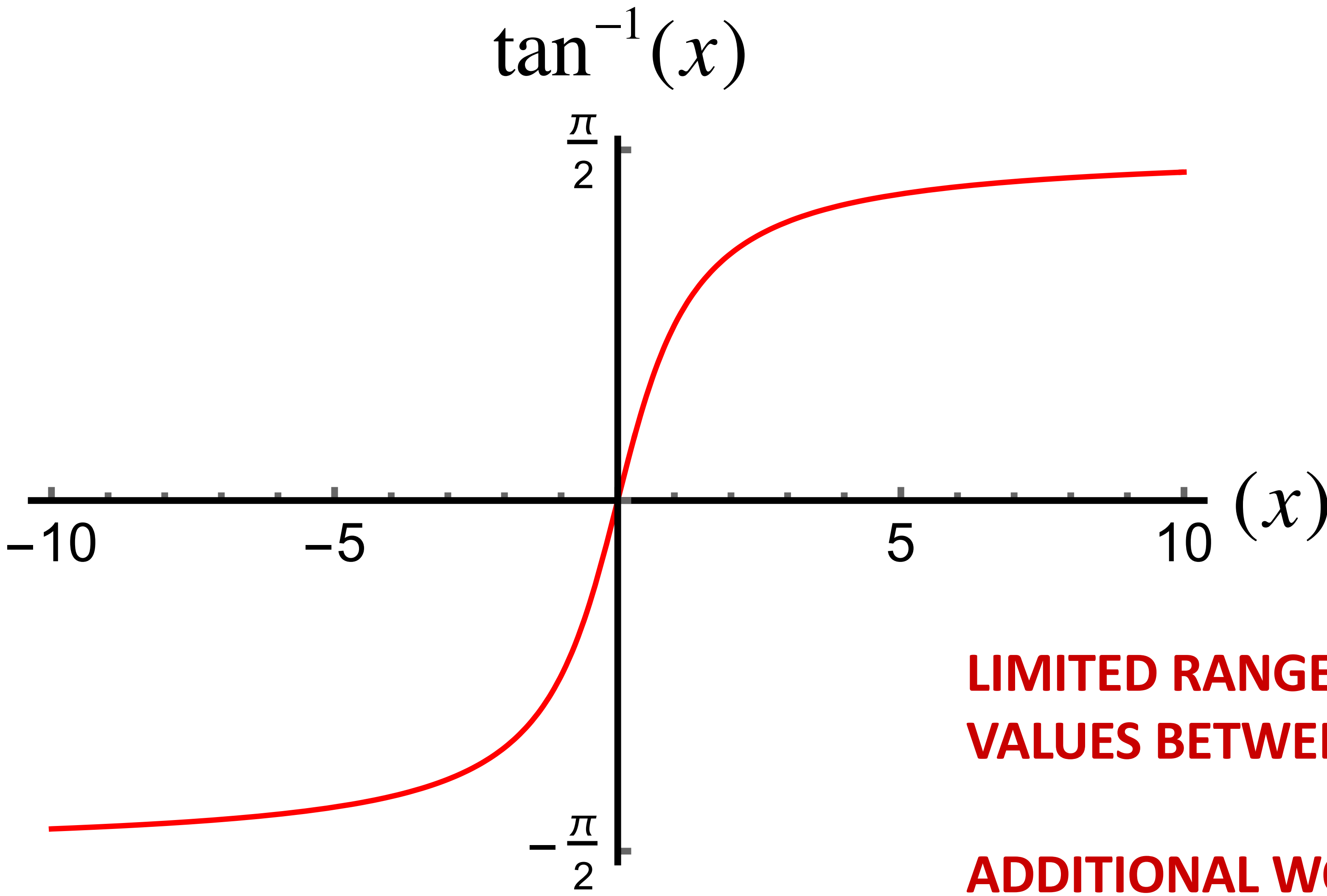
Recap: Tangent

A periodic function, but with a period π

– **Remember that cosine and sine have a period 2π**



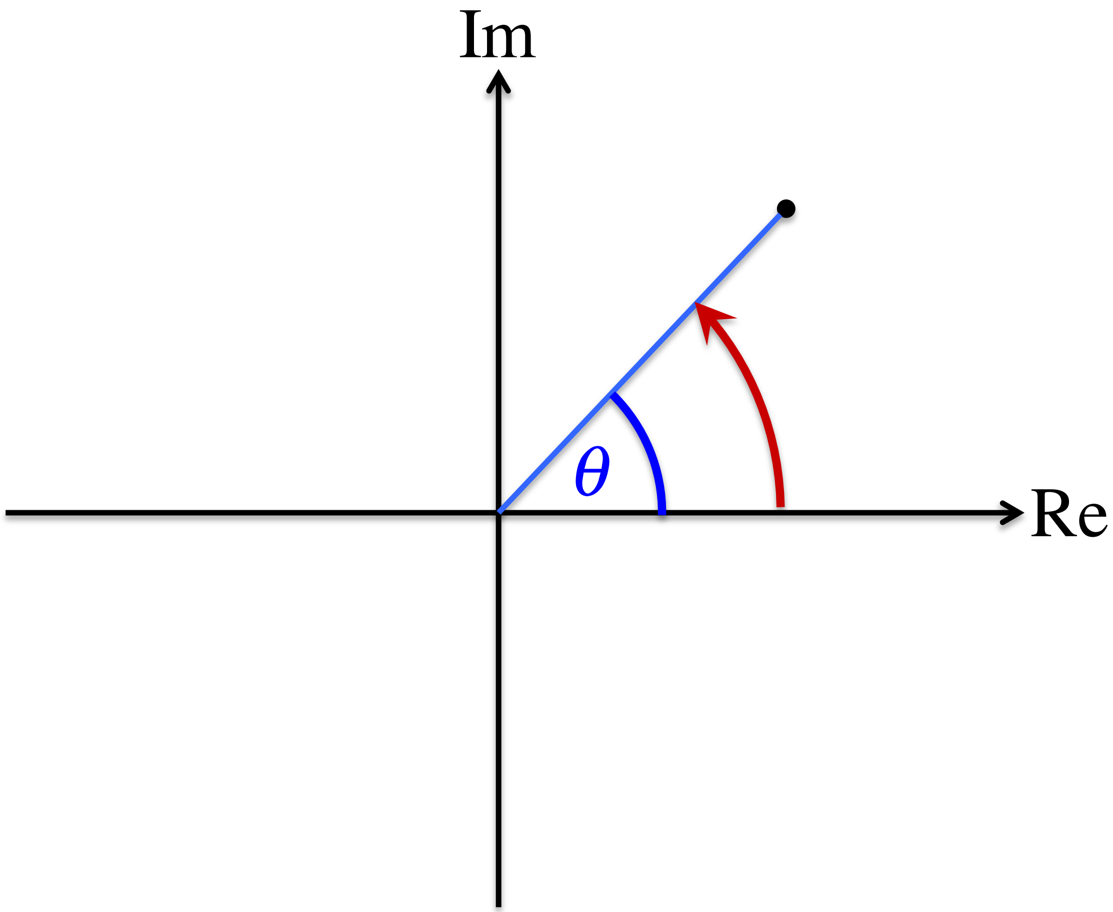
Inverse Tangent



LIMITED RANGE SINCE IT ONLY RETURNS VALUES BETWEEN - 90^{O} AND +90^{O}

ADDITIONAL WORK IS NEEDED TO GET THE CORRECT ANGLE FOR EACH QUADRANT!

Calculating the Angle



Case 1: $a_n > 0, \ b_n < 0 \Rightarrow -b_n > 0$

$$\theta = \tan^{-1}\left(\frac{-b_n}{a_n}\right)$$

$$\phi_n = \theta = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

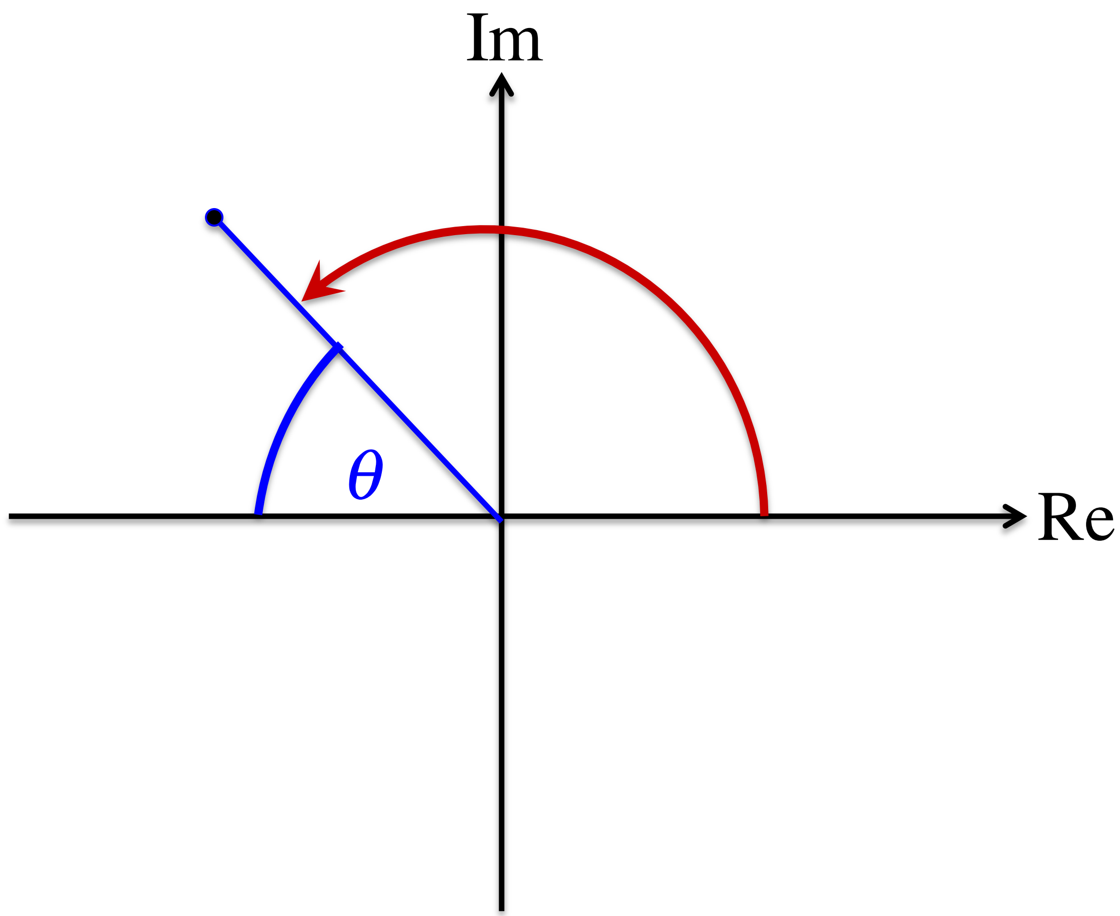
Note: We are plotting $(a_n, -b_n)$

Calculating the Angle

Case 2: $a_n < 0, \;\; b_n < 0 \Rightarrow -b_n > 0$

$$\theta = \tan^{-1}\left(\frac{-b_n}{-a_n}\right)$$

$$\phi_n = \pi - \theta = \pi - \tan^{-1}\left(\frac{b_n}{a_n}\right)$$



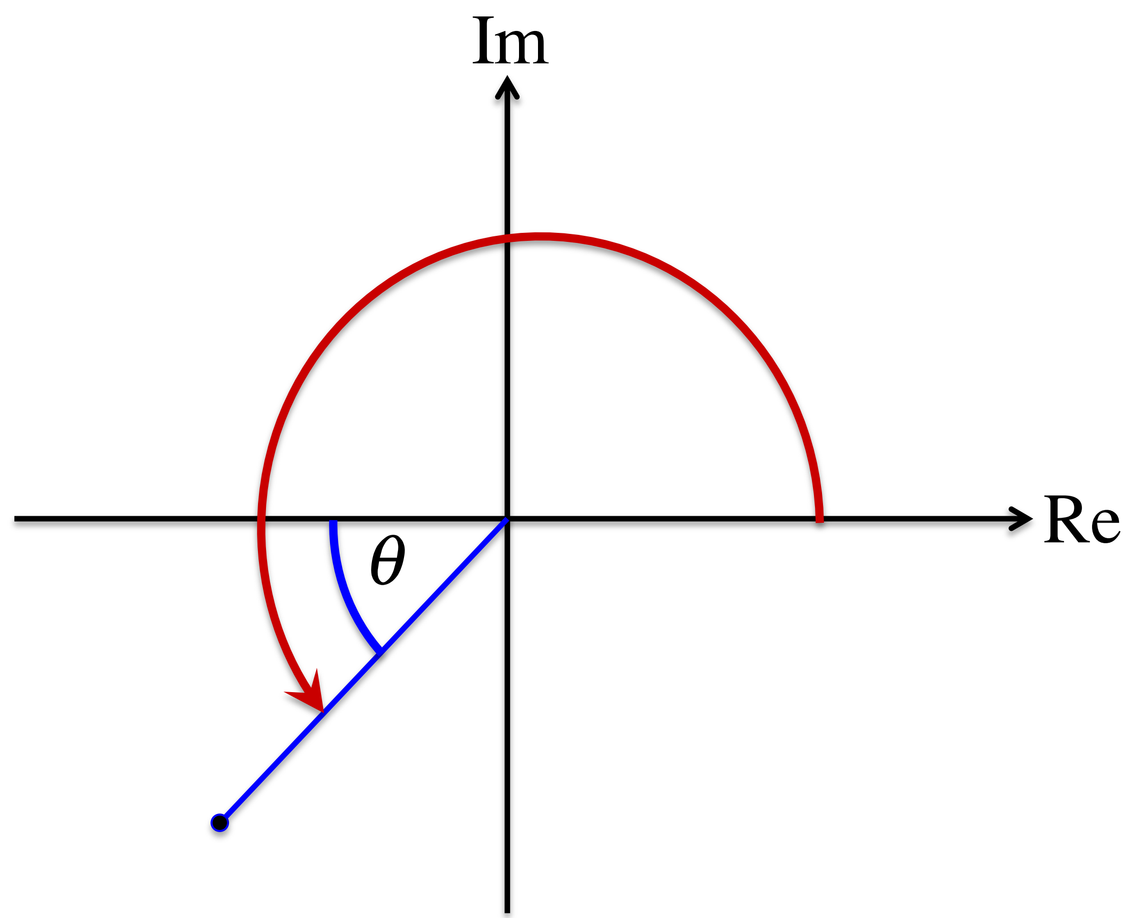
Note: We are plotting $(a_n, -b_n)$

Calculating the Angle

Case 3: $a_n < 0, \;\; b_n > 0 \Rightarrow -b_n < 0$

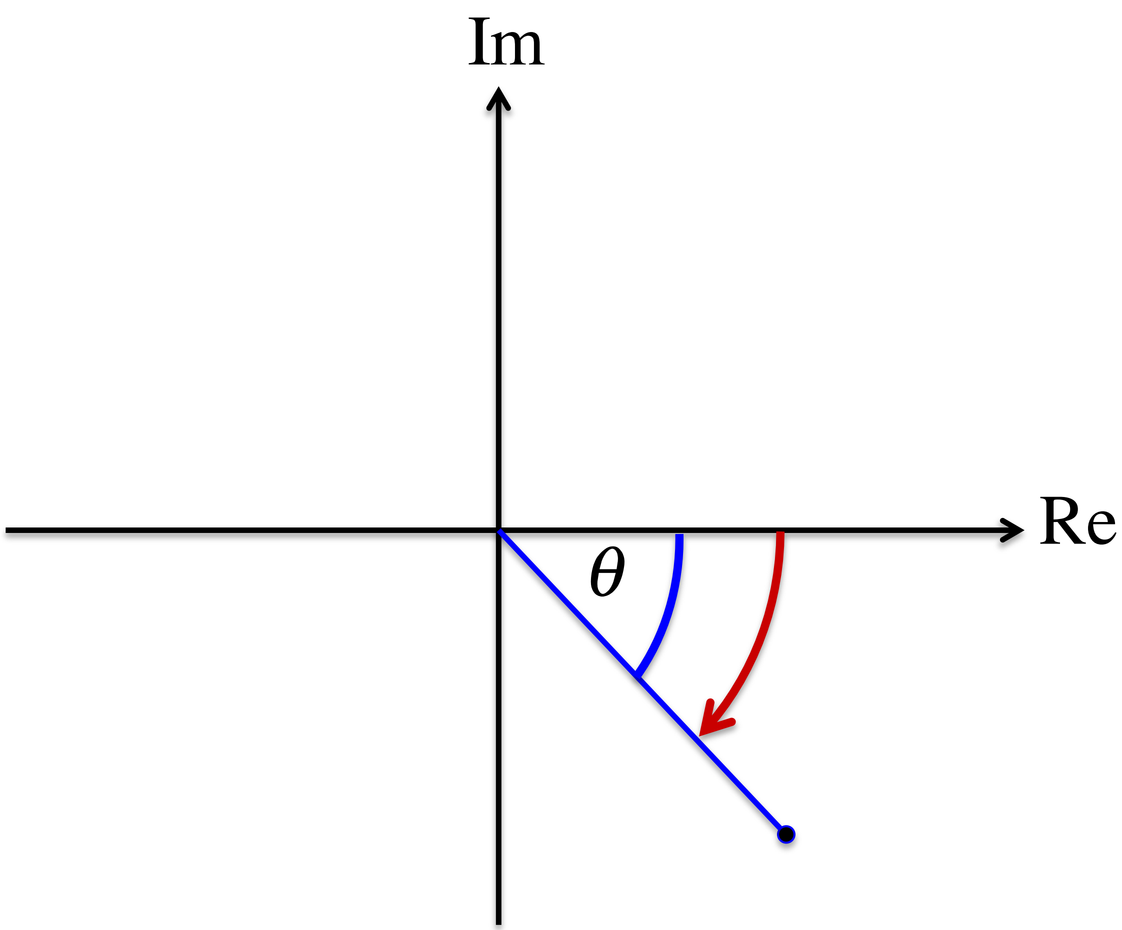
$$\theta = \tan^{-1}\left(\frac{b_n}{-a_n}\right)$$

$$\phi_n = \pi + \theta = \pi - \tan^{-1}\left(\frac{b_n}{a_n}\right)$$



Note: We are plotting $(a_n, -b_n)$

Calculating the Angle



Note: We are plotting $(a_n, -b_n)$

Case 4: $a_n > 0, b_n > 0 \Rightarrow -b_n < 0$

$$\theta = \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$\phi_n = -\theta = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

Common Formula

Case 2: $a_n < 0, b_n < 0$

$$\phi_n = \pi - \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

Case 1: $a_n > 0, b_n < 0$

$$\phi_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

$$\phi_n = \begin{cases} -\tan^{-1}\left(\dfrac{b_n}{a_n}\right), & a_n > 0 \\ \pi - \tan^{-1}\left(\dfrac{b_n}{a_n}\right), & a_n < 0 \end{cases}$$

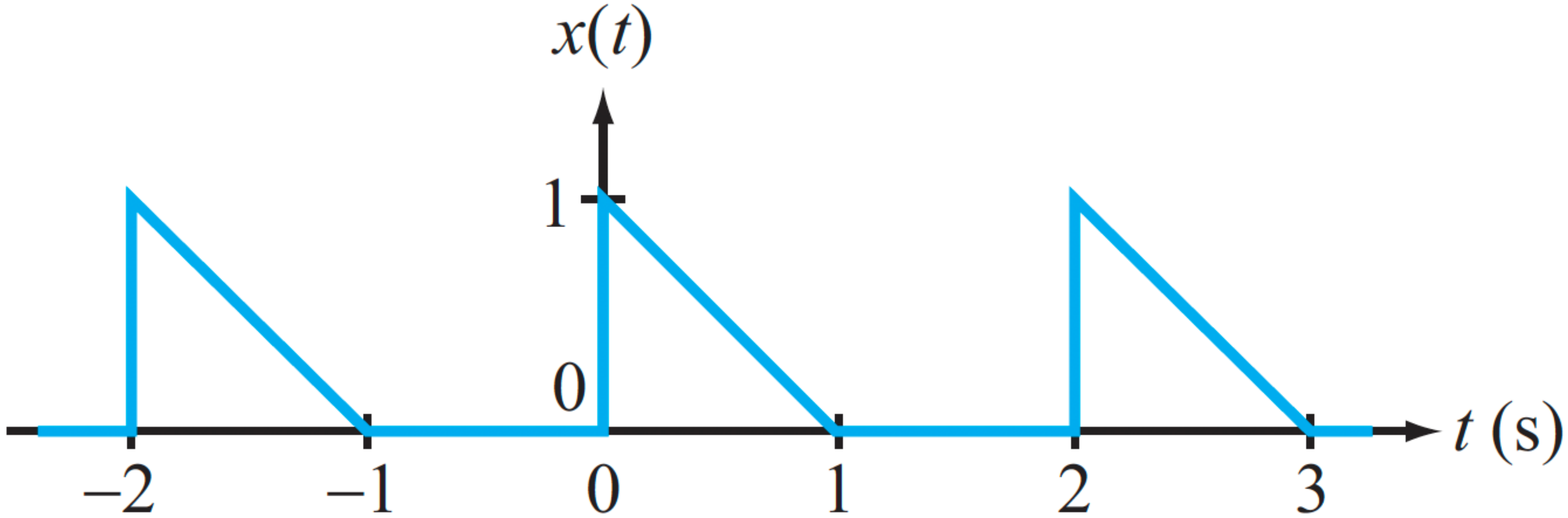
Case 3: $a_n < 0, b_n > 0$

$$\phi_n = \pi - \tan^{-1}\left(\frac{b_n}{a_n}\right)$$

Case 4: $a_n > 0, b_n > 0$

$$\phi_n = -\tan^{-1}\left(\frac{b_n}{a_n}\right)$$

Example



$$T_0 = 2$$

$$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt = \frac{1}{2}\int_0^1 (1-t)\,dt = 0.25$$

NO SYMMETRY, SO WE NEED TO CALCULATE ALL THE COEFFICIENTS

$$a_n = \frac{2}{T_0}\int_0^{T_0} x(t)\cos(n\omega_0 t)\,dt = \frac{2}{2}\int_0^1 (1-t)\cos(n\pi t)\,dt = \int_0^1 (1-t)\cos(n\pi t)\,dt$$

INTEGRATE BY PARTS

Recap: Integration by Parts

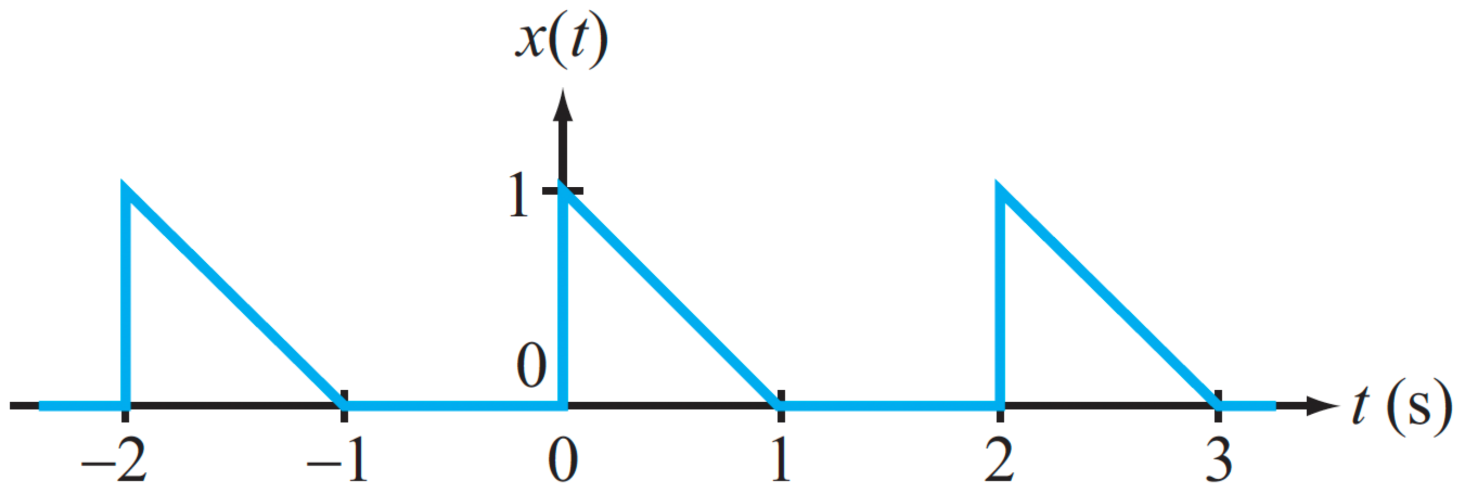
This is the basic method for simplifying the integral of a product of two or more functions

$$\int f(x)g(x)\,dx = f(x)G(x) - \int f'(x)G(x)\,dx$$

where:

$$G(x) = \int g(x)\,dx$$

Example



$$T_0 = 2$$

$$a_0 = \frac{1}{T_0}\int_0^{T_0} x(t)\,dt = \frac{1}{2}\int_0^1 (1-t)\,dt = 0.25$$

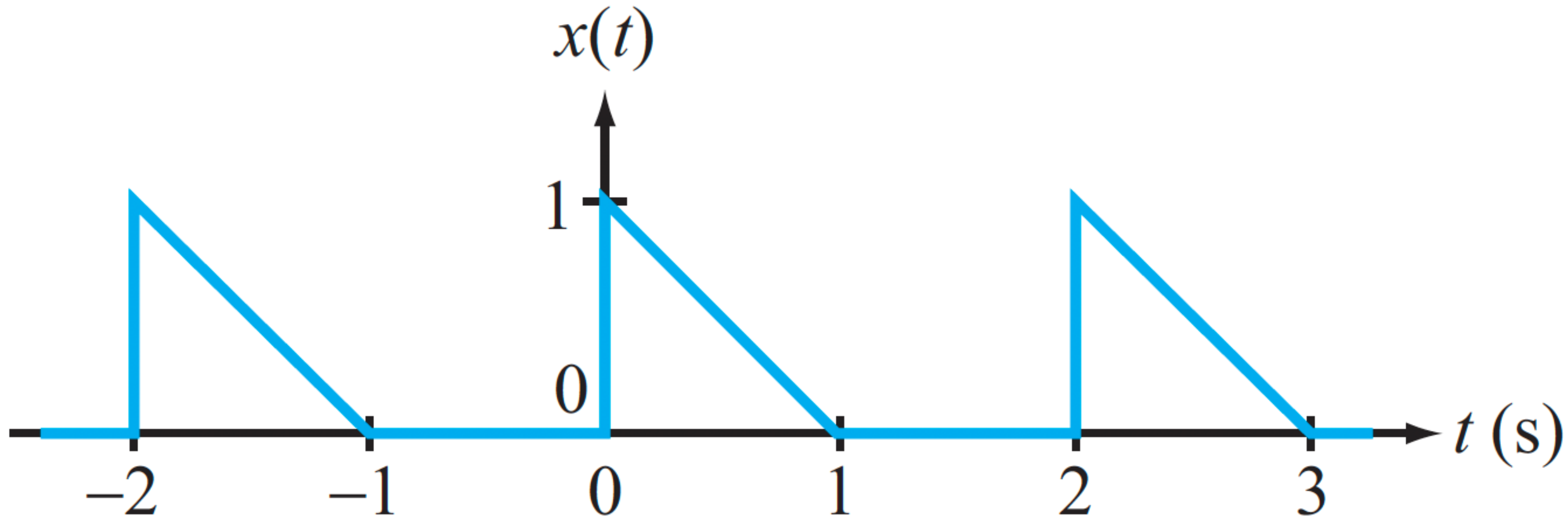
NO SYMMETRY, SO WE NEED TO CALCULATE ALL COEFFICIENTS

$$a_n = \int_0^1 (1-t)\cos(n\pi t)\,dt$$

INTEGRATE BY PARTS

$$= \cancel{(1-t)\frac{\sin(n\pi t)}{n\pi}\Bigg|_0^1} - \int_0^1 (-1)\frac{\sin(n\pi t)}{n\pi}\,dt = \frac{1}{n^2\pi^2}[1-\cos n\pi]$$

Example



$$b_n = \frac{2}{T_0}\int_0^{T_0} x(t)\sin(n\omega_0 t)\,dt = \int_0^1 (1-t)\sin(n\pi t)\,dt$$

$$= (1-t)\frac{-\cos(n\pi t)}{n\pi}\Bigg|_0^1 - \cancel{\int_0^1 (-1)\frac{(-\cos(n\pi t))}{n\pi}\,dt} = \frac{1}{n\pi}$$

INTEGRATE BY PARTS

Example (contd)

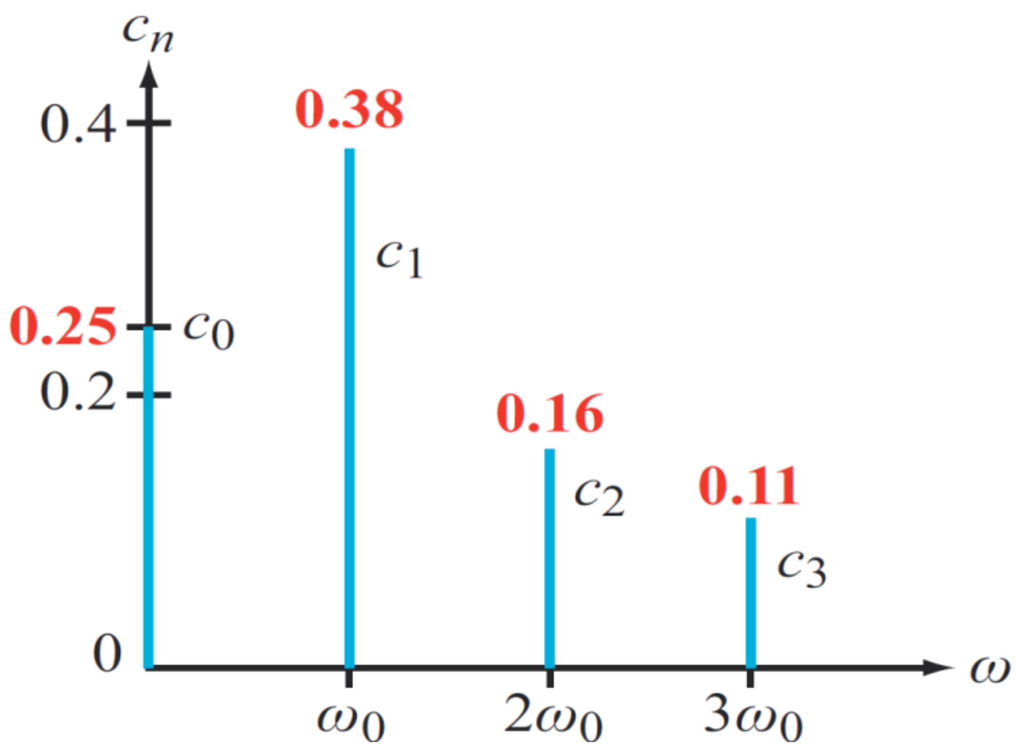
We can now compute the amplitude/phase version of the Fourier series:

$$c_n = \sqrt{a_n^2 + b_n^2} = \sqrt{\left(\frac{1}{n^2\pi^2}[1-\cos n\pi]\right)^2 + \frac{1}{n^2\pi^2}}$$

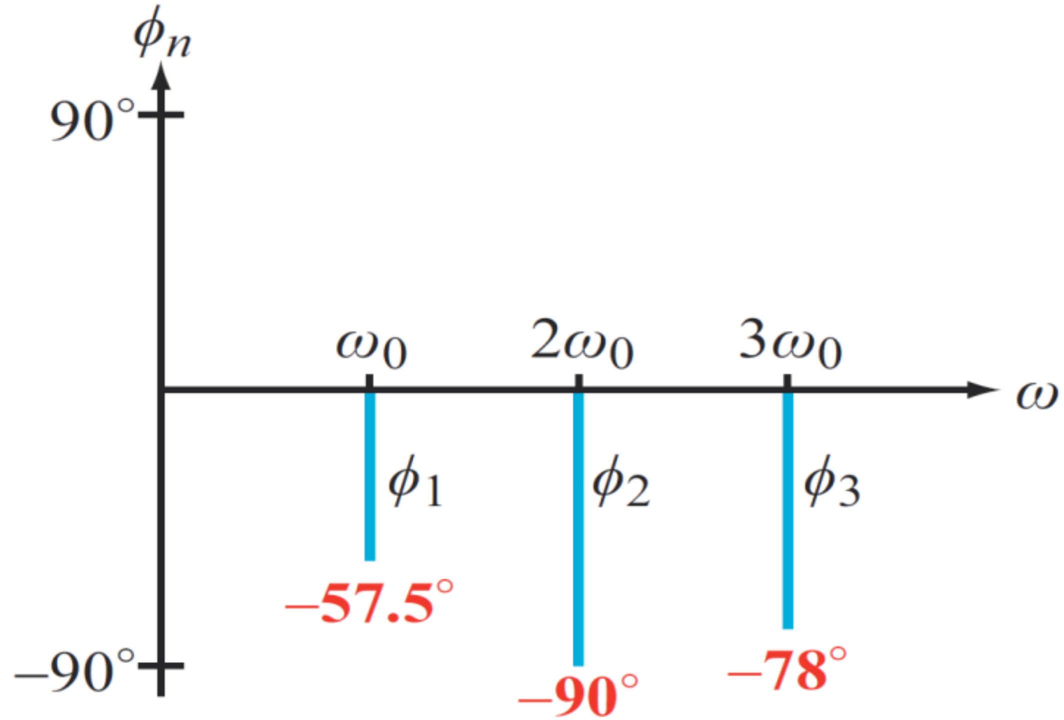
$$\phi_n = \tan^{-1}\left(\frac{-b_n}{a_n}\right) = -\tan^{-1}\left(\frac{n\pi}{[1-\cos n\pi]}\right)$$

These numbers are very informative – they tell us the magnitude and phase of each harmonic

Plotting the Fourier Coefficients



AMPLITUDE SPECTRUM



PHASE SPECTRUM

Spectra show "at a glance" which frequencies are most significant in the signal (bigger coefficients, more of the signal's energy), and which ones have less effect (smaller coefficients, less energy)

Three Forms of Fourier Series

$$x(t) = a_0 + \sum_{n=1}^{\infty} [a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]. \tag{5.27}$$

(sine/cosine representation)

✓

$$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n),$$

(amplitude/phase representation)

✓

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t},$$

(exponential representation)

NEXT

#3 Exponential Representation

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)\right]$$

$$\left.\begin{aligned}\cos(n\omega_0 t) &= \frac{1}{2}\left(e^{jn\omega_0 t} + e^{-jn\omega_0 t}\right)\\ \sin(n\omega_0 t) &= \frac{1}{2j}\left(e^{jn\omega_0 t} - e^{-jn\omega_0 t}\right)\end{aligned}\right\}$$

EULER'S FORMULAS

$$\Rightarrow x(t) = a_0 + \sum_{n=1}^{\infty}\left[\frac{a_n}{2}\left(e^{jn\omega_0 t} + e^{-jn\omega_0 t}\right) + \frac{b_n}{2j}\left(e^{jn\omega_0 t} - e^{-jn\omega_0 t}\right)\right]$$

$$= a_0 + \sum_{n=1}^{\infty}\left[\frac{(a_n - jb_n)}{2}e^{jn\omega_0 t} + \frac{(a_n + jb_n)}{2}e^{-jn\omega_0 t}\right]$$

Exponential Representation

$$x(t) = a_0 + \sum_{n=1}^{\infty}\left[\frac{(a_n - jb_n)}{2}e^{jn\omega_0 t} + \frac{(a_n + jb_n)}{2}e^{-jn\omega_0 t}\right]$$

$$= a_0 + \sum_{n=1}^{\infty}\left[\mathbf{x}_n e^{jn\omega_0 t} + \mathbf{x}_{-n} e^{-jn\omega_0 t}\right]$$

$$= \sum_{n=-\infty}^{\infty}\mathbf{x}_n e^{jn\omega_0 t}$$

$$\mathbf{x}_0 = a_0$$

$$\boxed{\mathbf{x}_{-n} = \frac{(a_n + jb_n)}{2} = \mathbf{x}_n^*}$$

CONJUGATE

THIS IS THE CLEANEST REPRESENTATION

Coefficients in the Exponential Representation

$$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$$

$$\Rightarrow \int_{-T_0/2}^{T_0/2} x(t)e^{-jm\omega_0 t}\,dt = \int_{-T_0/2}^{T_0/2} \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t} e^{-jm\omega_0 t}\,dt$$

Again, we see that when *n* = *m*, the exponential is 1.

$$= \sum_{n=-\infty}^{\infty} \int_{-T_0/2}^{T_0/2} \mathbf{x}_n e^{j(n-m)\omega_0 t}\,dt$$

For all other terms, the integral is zero

$$= T_0 \mathbf{x}_n$$

$$\boxed{\mathbf{x}_n = \frac{1}{T_0} \int_{-T_0/2}^{T_0/2} x(t)e^{-jn\omega_0 t}\,dt}$$

THE CLEANEST DERIVATION !

Recap: Symmetry

It is helpful to examine the waveform to see if it is odd or even symmetric.

- Sometimes, this is revealed when you remove the dc term
- Symmetry implies that one of the coefficients may be zero.

Even Symmetry: $x(t) = x(-t)$

$$a_0 = \frac{2}{T_0} \int_0^{T_0/2} x(t)\, dt$$

$$a_n = \frac{4}{T_0} \int_0^{T_0/2} x(t)\, \cos(n\omega_0 t)\, dt$$

$$b_n = 0$$

$$c_n = |a_n|, \qquad \phi_n = \begin{cases} 0 & \text{if } a_n > 0 \\ 180^\circ & \text{if } a_n < 0 \end{cases}$$

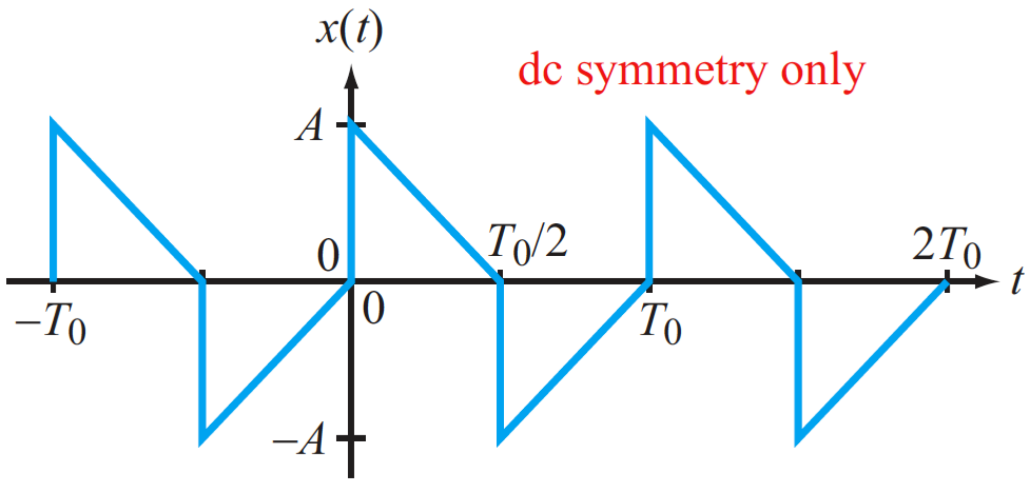
Odd Symmetry: $x(t) = -x(-t)$

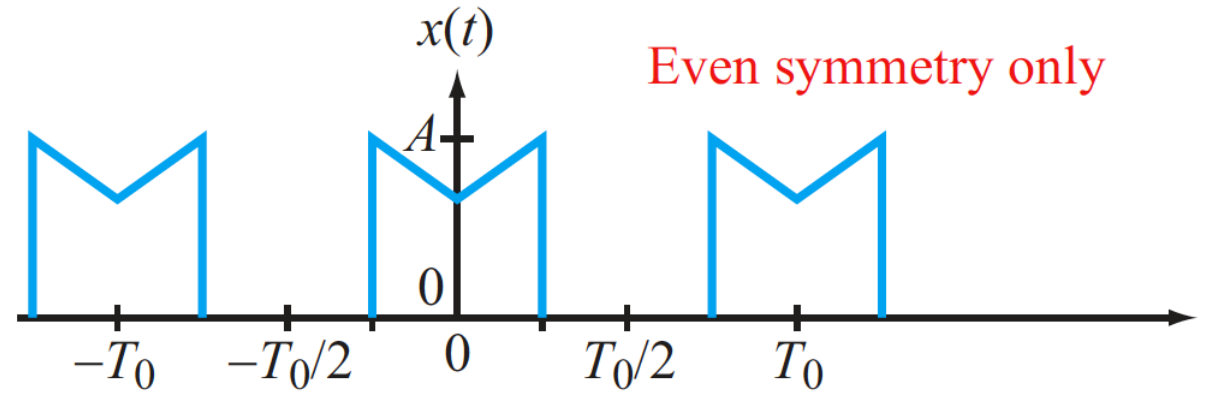
$$a_0 = 0 \qquad a_n = 0$$

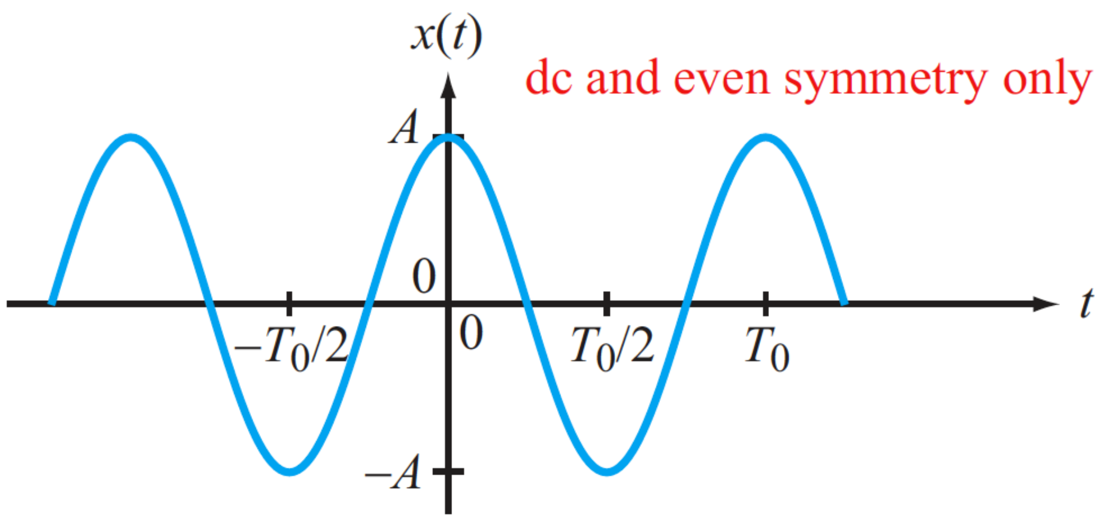
$$b_n = \frac{4}{T_0} \int_0^{T_0/2} x(t)\, \sin(n\omega_0 t)\, dt$$

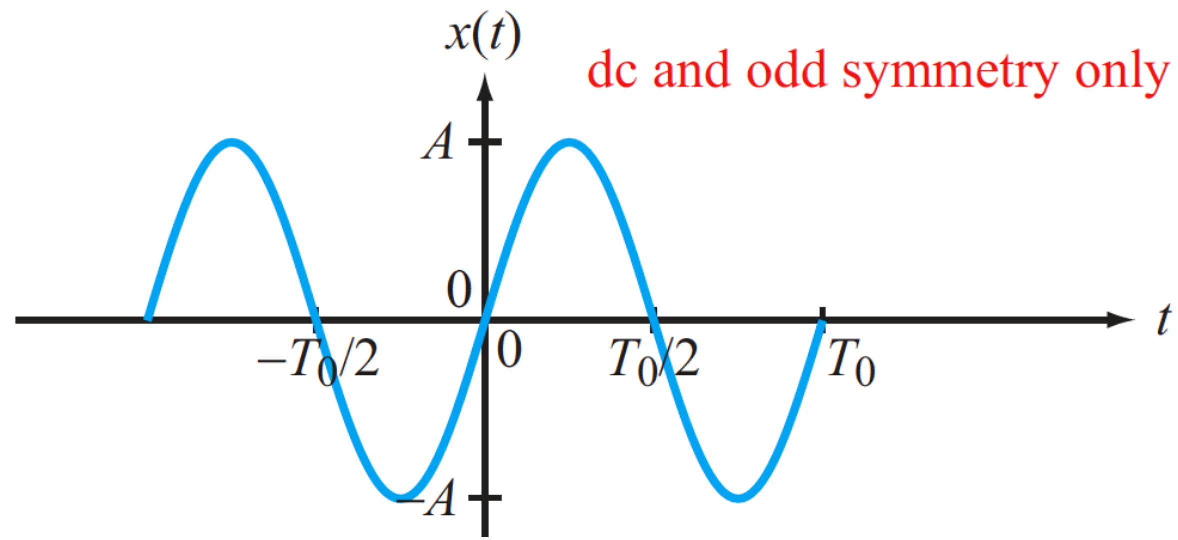
$$c_n = |b_n| \qquad \phi_n = \begin{cases} -90^\circ & \text{if } b_n > 0 \\ 90^\circ & \text{if } b_n < 0 \end{cases}$$

More Examples









Applying Signal Transformations

From Table 5-4:

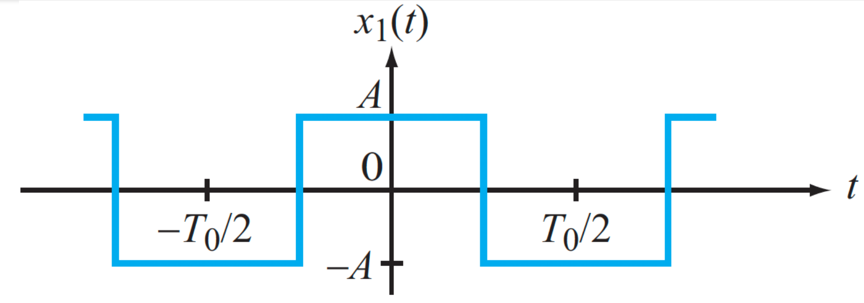
$$x_1(t) = \sum_{n=1}^{\infty} \frac{4A}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$$

$$x_2(t) = \frac{B}{2} + \left(\frac{B}{2A}\right) x_1(t)$$

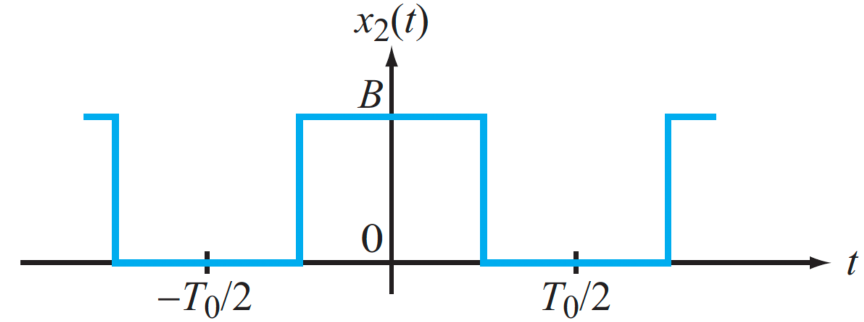
$$= \frac{B}{2} + \sum_{n=1}^{\infty} \frac{2B}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left(\frac{2n\pi t}{T_0}\right)$$

$$x_3(t) = x_1\left(t - \frac{T_0}{4}\right)$$

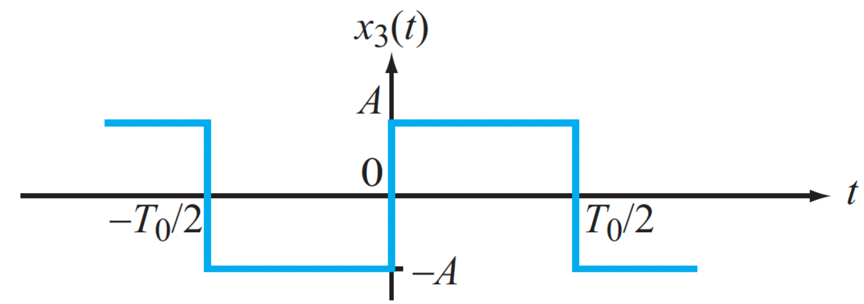
$$= \sum_{n=1}^{\infty} \frac{4A}{n\pi} \sin\left(\frac{n\pi}{2}\right) \cos\left[\frac{2n\pi}{T_0}\left(t - \frac{T_0}{4}\right)\right]$$



(a) $x_1(t)$



(b) $x_2(t)$

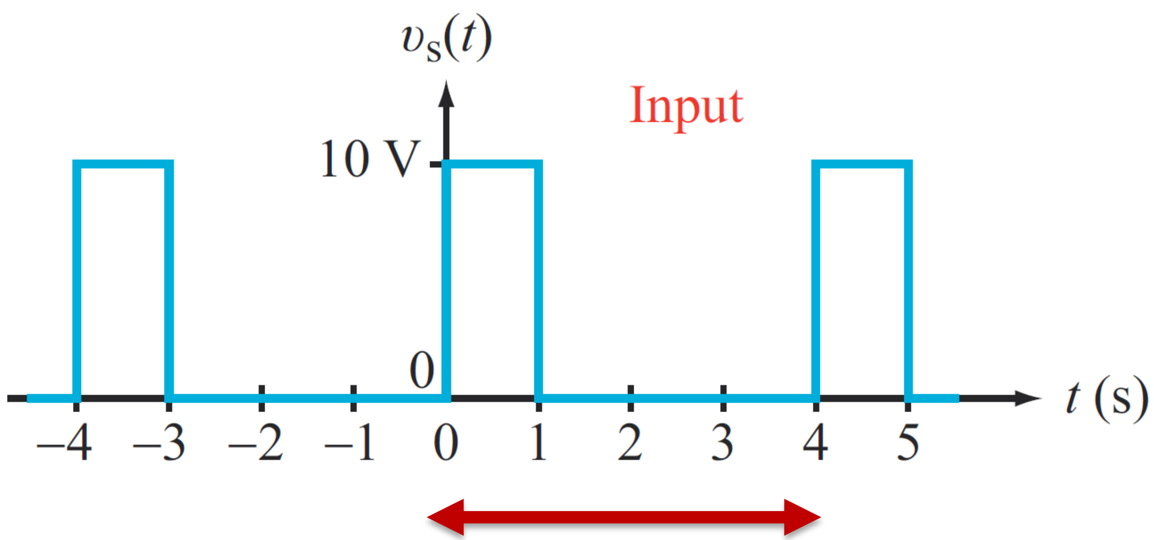


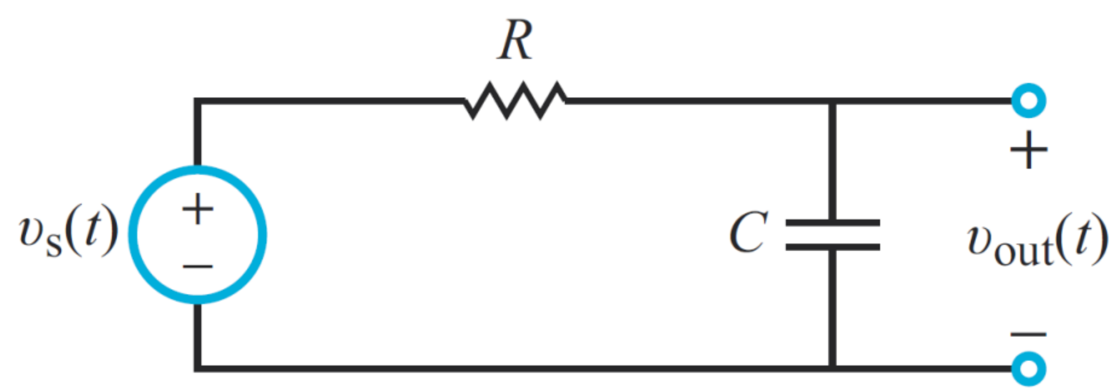
(c) $x_3(t)$

Circuit Analysis with Fourier Series

Compute the voltage response of the RC circuit shown to the input square wave shown, using Fourier series.

R = 20 kΩ, C = 0.1 mF, so RC = 2 sec





$$T_0 = 4 \quad \Rightarrow \omega_0 = \frac{2\pi}{4} = \frac{\pi}{2}$$

The Procedure

Step 1: Express $v_s(t)$ in terms of an amplitude/phase Fourier Series:

$$\upsilon_s(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n) \qquad (1)$$

Step 2: Determine the transfer function of the circuit at frequency *ω*:

$$\hat{\mathbf{H}}(\omega) = \mathbf{V}_{out} \quad \text{when } \upsilon_s = \cos\omega t$$

Step 3: Adjust each term in equation (1) by the transfer function from Step 2:

$$\upsilon_{out}(t) = a_0\hat{\mathbf{H}}(0) + \sum_{n=1}^{\infty} c_n \left|\hat{\mathbf{H}}\left(n\omega_0\right)\right| \cos(n\omega_0 + \phi_n + \angle\hat{\mathbf{H}}\left(n\omega_0\right))$$

Step 1: Fourier Representations

$$\upsilon_s(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$$

$$a_0 = \frac{1}{T_0}\int_0^{T_0} \upsilon_s(t)\,dt = \frac{1}{4}\int_0^{1} 10\,dt = 2.5$$

$$a_n = \frac{2}{4}\int_0^{1} \upsilon_s(t)\cos\left(\frac{n\pi t}{2}\right)dt = \frac{10}{n\pi}\sin\frac{n\pi}{2}$$

$$b_n = \frac{2}{4}\int_0^{1} \upsilon_s(t)\sin\left(\frac{n\pi t}{2}\right)dt = \frac{10}{n\pi}\left(1 - \cos\frac{n\pi}{2}\right)$$

Step 1: Fourier Representations

$$a_n = \frac{2}{4}\int_0^1 \upsilon_s(t)\cos\left(\frac{n\pi t}{2}\right)dt = \frac{10}{n\pi}\sin\frac{n\pi}{2}$$

$$b_n = \frac{2}{4}\int_0^1 \upsilon_s(t)\sin\left(\frac{n\pi t}{2}\right)dt = \frac{10}{n\pi}\left(1-\cos\frac{n\pi}{2}\right)$$

$$\Rightarrow c_n e^{j\phi_n} = a_n - jb_n = \frac{10}{n\pi}\left[\sin\frac{n\pi}{2} - j\left(1-\cos\frac{n\pi}{2}\right)\right]$$

What This Means

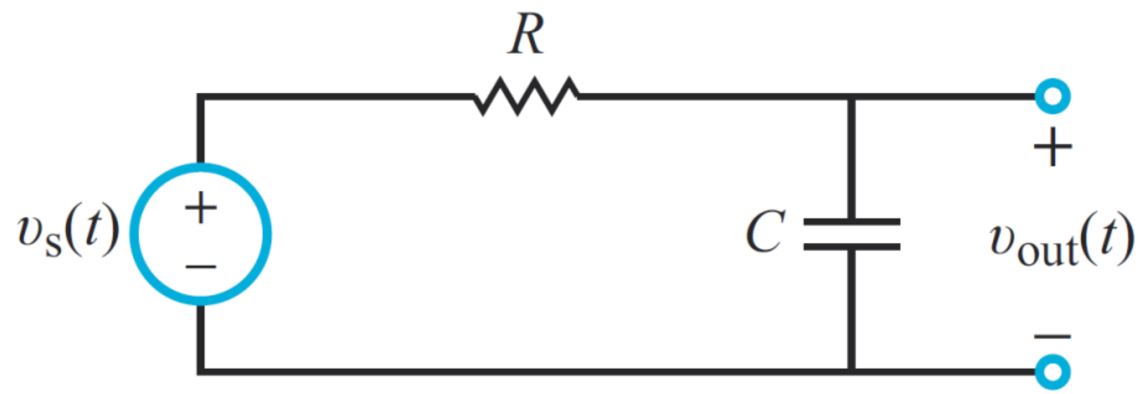
$$v_s(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$$

- When this signal goes through the circuit with transfer function $\hat{\mathbf{H}}(\omega)$, each term is altered in a frequency dependent fashion
- The magnitude of $c_n \cos(n\omega_0 + \phi_n)$ is multiplied by $|\hat{\mathbf{H}}(n\omega_0)|$
- The phase $c_n \cos(n\omega_0 + \phi_n)$ is adjusted by $\angle\hat{\mathbf{H}}(n\omega_0)$

Step 2: $\hat{\mathbf{H}}(\omega)$

$$\hat{\mathbf{H}}(\omega) = \frac{1}{1 + j\omega RC}$$

VOLTAGE DIVIDER



$$|\hat{\mathbf{H}}(\omega)| = \frac{1}{\sqrt{1+\omega^2 R^2 C^2}} = \underline{\frac{1}{\sqrt{1+4\omega^2}}}$$

R = 20 kΩ,

C = 0.1 mF, so

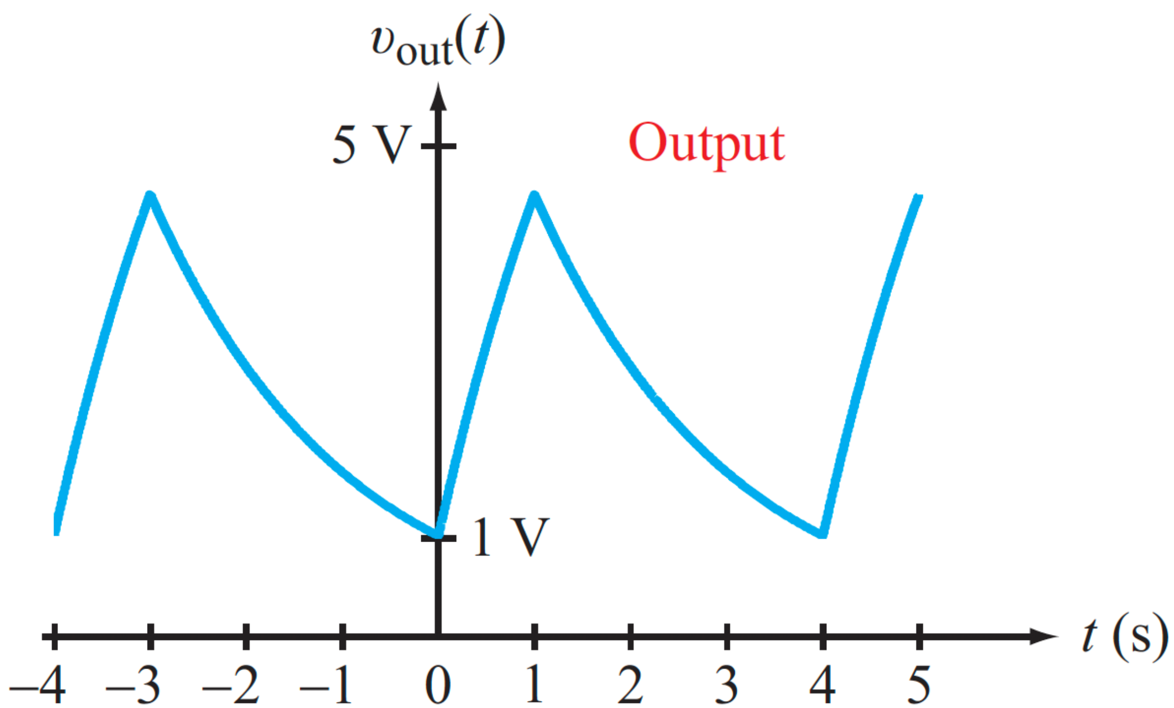
RC = 2 sec

$$\angle\hat{\mathbf{H}}(\omega) = -\tan^{-1}(\omega RC) = \underline{-\tan^{-1}(2\omega)}$$

Step 3: Output Waveform

$$v_{out}(t) = 2.5 + \sum_{n=1}^{\infty} c_n \frac{1}{\sqrt{1+4n^2\omega_0^{\,2}}} \cos(n\omega_0 t + \phi_n - \tan^{-1}(2n\omega_0))$$

We will skip the numerical calculations in class, (requires computer) but the results look like this:



Summary: Three Forms of the Fourier Series

Cosine/Sine	Amplitude/Phase	Exponential
$x(t) = a_0 + \sum_{n=1}^{\infty}[a_n \cos(n\omega_0 t) + b_n \sin(n\omega_0 t)]$	$x(t) = c_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n)$	$x(t) = \sum_{n=-\infty}^{\infty} \mathbf{x}_n e^{jn\omega_0 t}$
$a_0 = \frac{1}{T_0} \int_0^{T_0} x(t)\, dt$	$c_n e^{j\phi_n} = a_n - jb_n$	$\mathbf{x}_n = \|\mathbf{x}_n\| e^{j\phi_n};\ \mathbf{x}_{-n} = \mathbf{x}_n^*;\ \phi_{-n} = -\phi_n$
$a_n = \frac{2}{T_0} \int_0^{T_0} x(t)\ \cos n\omega_0 t\ dt$	$c_n = \sqrt{a_n^2 + b_n^2}$	$\|\mathbf{x}_n\| = c_n/2;\ x_0 = c_0$
$b_n = \frac{2}{T_0} \int_0^{T_0} x(t)\ \sin n\omega_0 t\ dt$	$\phi_n = \begin{cases} -\tan^{-1}(b_n/a_n), & a_n > 0 \\ \pi - \tan^{-1}(b_n/a_n), & a_n < 0 \end{cases}$	$\mathbf{x}_n = \frac{1}{T_0} \int_0^{T_0} x(t)\ e^{-jn\omega_0 t}\ dt$

$$a_0 = c_0 = x_0;\ a_n = c_n \cos\phi_n;\ b_n = -c_n \sin\phi_n;\ \mathbf{x}_n = \tfrac{1}{2}(a_n - jb_n)$$

Summary: Application to Circuits

Step 1: Express $v_s(t)$ in terms of an amplitude/phase Fourier Series:

$$\upsilon_s(t) = a_0 + \sum_{n=1}^{\infty} c_n \cos(n\omega_0 t + \phi_n) \qquad (1)$$

Step 2: Determine the transfer function of the circuit at frequency *ω:*

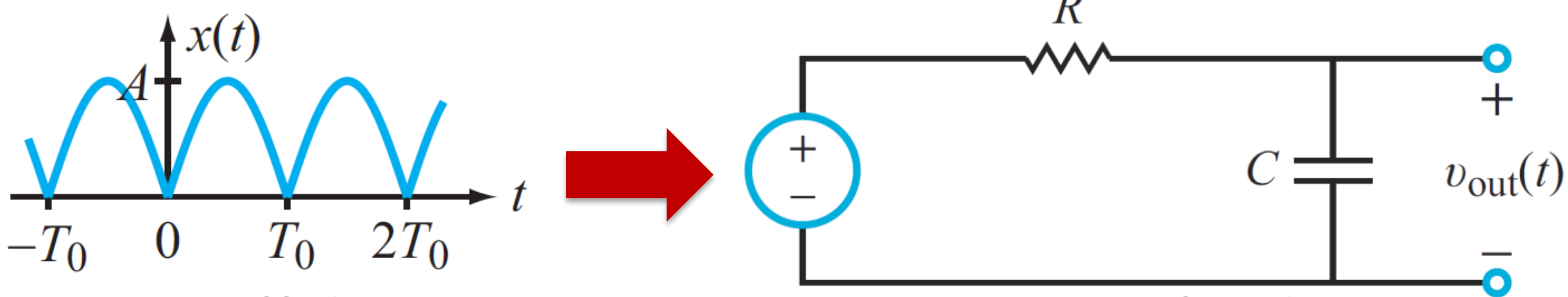
$$\hat{\mathbf{H}}(\omega) = \mathbf{V}_{out} \quad \text{when } \upsilon_s = \cos \omega t$$

Step 3: Adjust each term in equation (1) by the transfer function from Step 2:

$$\upsilon_{out}(t) = a_0 \hat{\mathbf{H}}(0) + \sum_{n=1}^{\infty} c_n \left|\hat{\mathbf{H}}\left(n\omega_0\right)\right| \cos(n\omega_0 + \phi_n + \angle\hat{\mathbf{H}}\left(n\omega_0\right))$$

Activity

- **Write down a formula for the output of the RC circuit when the input is the full-wave rectified sinusoid (Table 5-4, entry #8)**



- **Step 1: Read off the Fourier series representation for the input waveform from Table 5-4:**

$$x(t) = \frac{2A}{\pi} + \sum_{n=1}^{\infty} \frac{4A}{\pi(1-4n^2)} \cos\left(\frac{2n\pi t}{T_0}\right)$$

- **Step 2: Write down the transfer function for the RC circuit** $\hat{\mathrm{H}}(\omega) = \dfrac{1}{1 + j\omega RC}$
- **Step 3: Adjust each term of the Fourier series from Step 1 using the transfer function in Step 2 to write the answer !**

For the Next Class

Work through the textbook examples in Section 5-4.3 – 5-5

Study Sections 5.6 – 5.7

<u>How to Study</u>:

1. **Read the assigned sections in order to understand the basic concepts**
2. **Memorize the concepts**
3. **Attempt the in-chapter exercises and practice until you can do them fully on your own without notes, like in a test**
4. **Write down your questions and bring them to class**